THE UNIVERSITY OF HONG KONG

BACHELOR OF ENGINEERING EXAMINATION

DEPARTMENT OF CIVIL ENGINEERING

CIVL2008 / CIVL2108: PRINCIPLES OF CIVIL ENGINEERING MANAGEMENT

December 12, 2014 2:30 pm – 5:30 pm (3 hours)

Answer **ANY FOUR** questions

All questions carry equal marks

Use of Calculators

Only approved calculators as announced by the Examinations Secretary can be used in this examination. It is candidates' responsibility to ensure that their calculator operates satisfactorily, and candidates must record the name and type of the calculator used on the front page of the examination script.

1. a) As a strong supporter of sustainable development, a private developer is keen to bring in more advanced design and technological solutions to their new residential development schemes. Discuss whether the design and build procurement method can maximise the opportunity for innovation in this type of projects. [12 marks]

   b) Explain with detailed justifications other procurement method(s) to be considered by the developer if their company goal is to be a market leader in sustainable development in Hong Kong. [13 marks]

2. a) When examining a tender received, the engineer has discovered that the tender price has been intentionally front loaded and the tender also contains certain qualifications. Explain with examples what are 'front loading' and 'qualification' in tendering sense and how should the engineer deal with these tendering issues? [8 marks]

   b) If arithmetical errors exist in a lump sum tender, describe the detailed procedures to be followed to handle arithmetical errors and explain from the client's perspective why such procedures are needed? [12 marks]

   c) Explain what is a 'covered price' tender and discuss whether it is ethical for the contractor to submit a tender with a covered price? [5 marks]

3. a) A consortium is considering submitting a proposal for the construction and operation of a toll road. The project will cost $2 billion to build. It is anticipated that 30,000 vehicles will be commuting through this toll road every day at a charge of $30 per vehicle per journey. However, major preventive maintenance programme at the cost of $0.2 billion shall be conducted at a six-year interval. As for the interest charge as well as the operation and routine maintenance costs, they are estimated at $0.05 billion per year and $0.03 billion per year respectively. The toll road will be returned to the government at no charge after 15 years of operation by the consortium. Calculate the internal rate of return for this scheme and comment on its financial viability. [18 marks]

   b) Highlight the major risks associated with the project described in (a) and explain how to avoid or minimise those risks from occurring. [7 marks]

4. a) Explain with relevant examples why major infrastructure projects are more susceptible to the risks of delay, cost overrun and poor quality standard. [10 marks]

   b) Describe the roles of project managers in different project stages and discuss how project managers can help ensure the project objectives are satisfactorily achieved. [15 marks]

5. a) The schedule and logic of a small-scale construction project are shown in the following table:

Activity	*Immediately Preceding Activity*	*Duration*
A	--	4
B	A	12
C	A	20
D	A	8
E	B	14
F	B, C	10
G	C, D	6
H	E, F	10
I	G, H	4

i) Prepare an "activity-on-node" network diagram using the above information and indicate clearly the critical path of this network. [6 marks]

ii) Compute the total float and free float for each activity based on (i) above. [6 marks]

iii) Explain why there could be a difference between the total float and free float and justify the need for calculating the free float from the planning perspective. [5 marks]

b) With the aid of relevant diagrams, describe the way in which repetitive works should be planned so as to avoid labour and equipment from staying idle. Explain how to work out the 'turnover' when preparing a repetitive programme. [8 marks]

~ END OF PAPER ~